Final Design Doc

Product: Short form video analysis system that will take an input in the form of a transcript scraped from YouTube from a short form video, along with the video's description, title, and audio. and will then output a bias lean for the video using a bias measurement system from a standard source. Web browser extension that helps users think critically while scrolling.

MVP: Select one Youtube video from a verified and popular news source. A model takes in data scraped from the video link, and outputs the biasedness.

Backend:

1. Scrape data from youtube with **Apify,** returns a clean JSON object containing the transcript string, title, description, and uses yt-dlp to get and transmit video audio.
2. Fact check data with bright data, or Tavily api (if bright data does not prove suitable). Conducts a 1-step web search to check the factual claims made in the transcript before calculating bias.
3. Processing the data. Plan A: use an LLM. Processes audio and text in a single pass to evaluate: Political lean, Reliability, Sensationalism/tone, Ownership & Transparency
4. Output: Ratings formatted strictly on the AllSides 5-Point Spectrum (*Left, Lean Left, Center, Lean Right, Right*), along with a short note summary on the main reason for the rating.

Frontend:

1. Youtube link uploaded to mana.space frontend website (keep it simple; just an input form, we are testing the backend with this project)-> call to backend
2. Send the final results from processed data to display in mana.space frontend as a small browser popup window.

Validation:
We will check manually if the video content matches the output of the product.

Additional Features:

1. Analyze videos as well instead of just shorts
2. Instead of using a link to video, scrape data from youtube in real time while the user is scrolling.
3. Generalize video content to not just politics.
4. Add more details to the output instead of just the allsides spectrum.

Risks and mitigation:

The LLM rater carries its own lean ⇒Test on videos from both sides; compare with a human panel before trusting it

Labels harm a named creator⇒Descriptive wording tied to quotes, no verdicts like "misinformation"; a way to flag errors and respond